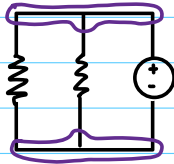
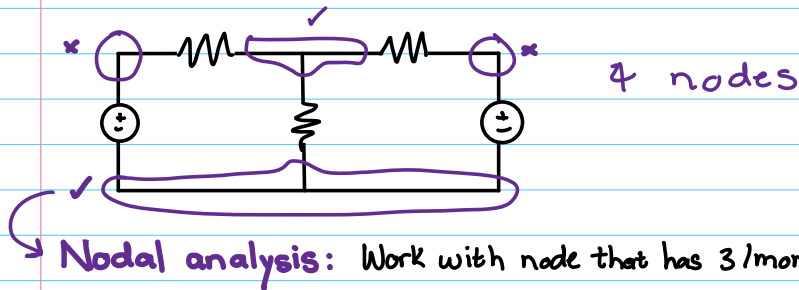


Uncle Madel - Tutor Class 3

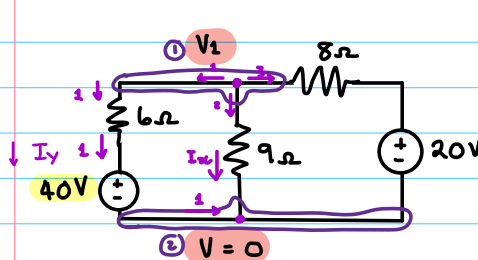
15 August 2022: 20:00 - 22:00

CHAPTER 3: Nodal analysis



1. Identify a node (where 3/more lines connecting)
2. Assign variable(s) & 1 node must sacrifice (has to be ground)
3. — (will do later)
4. Assume current is leaving the node that we are dealing with
5. Apply node analysis

EXAMPLE 1



1. ① + ②
2. let ① = V_1 + let ② be ground
let ② = V $\therefore V = 0$
- 3.
4. Ignore arrow: take current leaving
5. a) $I = \frac{V_{diff}}{R_{tot}}$ → where current coming from through all voltages to where going

$$V_1 - \boxed{40} - V$$

$$= V_1 - (+40) - 0$$

$$= V_1 - 40$$

*take out common factor:

or * use calculator

$$\frac{V_1 - 40}{6} + \frac{V_1 - 0 - 0}{9} + \frac{V_1 - 20 - 0}{8} = 0$$

$$\frac{V_1 + 40}{6} + \frac{V_1}{9} + \frac{V_1 - 20}{8} = 0$$

$$\frac{V_1}{6} - \frac{40}{6} + \frac{V_1}{9} + \frac{V_1}{8} - \frac{20}{8} = 0$$

$$V_1 \left(\frac{1}{6} + \frac{1}{9} + \frac{1}{8} \right) = \frac{40}{6} + \frac{20}{8}$$

$$V_1 = 22.76 \text{ V}$$

- Calculate I_x ↓: I_x same direction as I_2 ($\therefore$ agreeing our assumption from 4.)
 $\therefore I_x = \frac{V_1}{9} = \frac{22.76}{9} = 2.528 \text{ A}$

* If I_x was ↑: I_x in opposite direction as I_2 ($\therefore$ opposing our assumption from 4.)

$$\therefore I_x = -\frac{V_1}{9} = -\frac{22.76}{9} = -2.528 \text{ A}$$

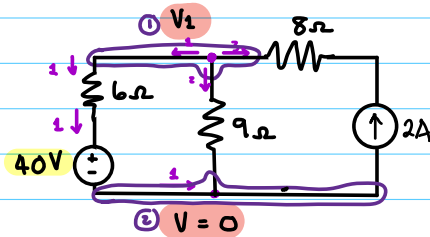
- Calculate $I \downarrow$: I_x same direction as I_1 ($\therefore$ agreeing our assumption from 4.)

$$\therefore I_x = \frac{V_1 - 40}{6} = \frac{22.76 - 40}{6} = -\Delta$$

- * If I was $\uparrow$: I_x in opposite direction as I_1 ($\therefore$ opposing our assumption from 4.)

$$\therefore I_x = \frac{-V_1 - 40}{6} = \frac{-22.76 - 40}{6} = -\Delta$$

EXAMPLE 2



1. ① + ②

2. let ① = V_1 & let ③ be ground
let ② = V $\therefore V = 0$

- 3.

4. Ignore arrow: take current leaving

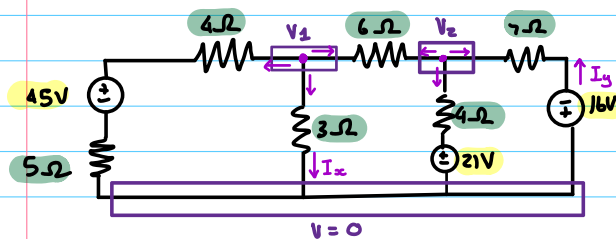
5. a) $I = \frac{V_{diff}}{R_{tot}}$ → where current coming from through all voltages to where going

$$\frac{V_1 - 40}{6} + \frac{V_1 - 0}{9} - 2 = 0$$

$$\frac{V_1 - 40}{6} + \frac{V_1}{9} - 2 = 0$$

$$V_1 = 31.2V$$

EXAMPLE 3



5. V_1 : $\leftarrow \downarrow \rightarrow$

$$\frac{V_1 - 45}{9} + \frac{V_1}{3} + \frac{V_1 - V_2}{6} = 0 \quad \# 2 \text{ variables}$$

$$\frac{V_1}{9} - 5 + \frac{V_1}{3} + \frac{V_1}{6} - \frac{V_2}{6} = 0$$

$$V_1 \left(\frac{1}{9} + \frac{1}{3} + \frac{1}{6} \right) - \frac{V_2}{6} = 5$$

$$\frac{11}{18} V_1 - \frac{1}{6} V_2 = 5 \quad \dots \dots \dots ①$$

- V_2 : $\leftarrow \downarrow \rightarrow$

$$\frac{V_2 - V_1}{6} + \frac{V_2 - 21}{4} + \frac{V_2 + 16}{7} = 0$$

$$V_2 \left(\frac{1}{6} + \frac{1}{4} + \frac{1}{7} \right) + V_1 \left(-\frac{1}{6} \right) - \frac{21}{4} + \frac{16}{7} = 0$$

$$-\frac{1}{6} V_1 + \frac{47}{84} V_2 = \frac{83}{28} \quad \dots \dots \dots ②$$

* Calculator: Mode $\rightarrow$ EQN $\rightarrow a_n X + b_n Y = c_n$

$$\therefore X = V_1 = 10.48$$

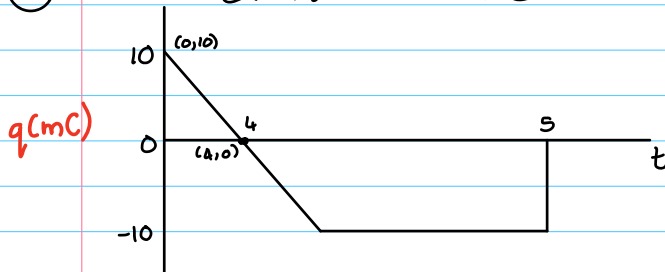
$$Y = V_2 = 8.42$$

Uncle Madel - Class Test 1 Prep

16 August 2022: 22:00 - 00:00

①

Calculate current (I) @ $t=4$



@ $t=4$

$$I = \frac{10-0}{0-4} = \frac{-10}{-4} \text{ A}$$

②

A 15Ω resistor is connected to a 10V battery.

How much energy will be dissipated in the resistor in 1 minute?

$R = 15\Omega$

$V = 10\text{V}$

$t = 1\text{min.} \approx 60\text{s}$

$P = \frac{W}{t}$ and $P = VI$

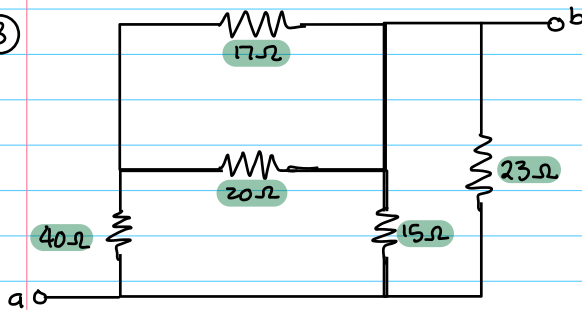
$W = P \cdot t$

$W = \left(\frac{V^2}{R}\right) \cdot t$

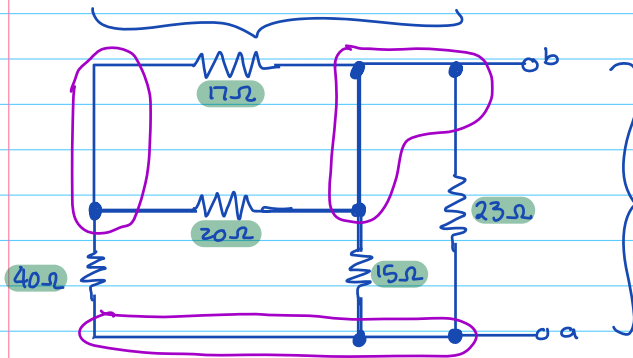
$W = \left(\frac{10^2}{15}\right)(60)$

$W = 400 \text{ J}$

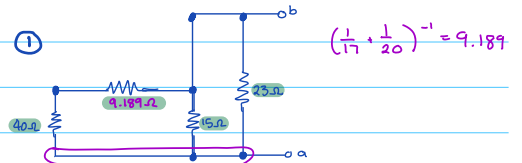
③



Calculate R_{eq} with respect to a-b

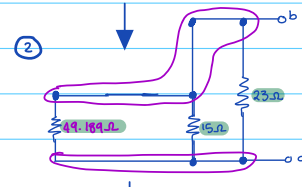


①

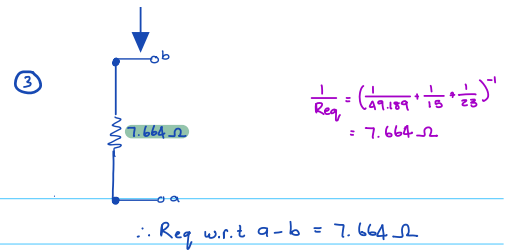


$\left(\frac{1}{17} + \frac{1}{20}\right)^{-1} = 9.189$

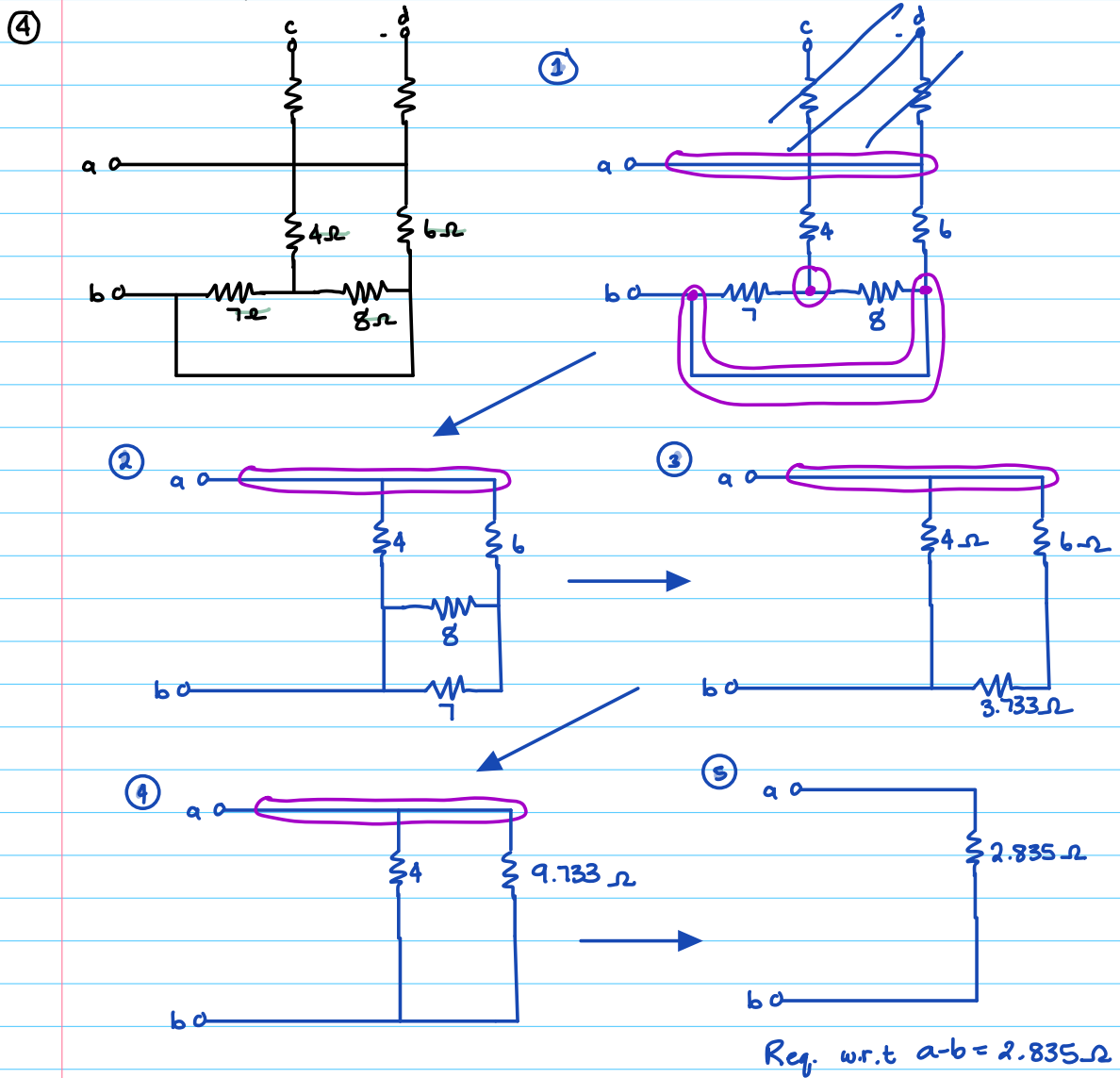
②



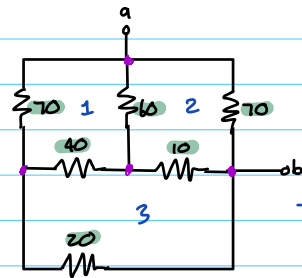
$R_{eq} = 40 + 9.189\Omega$



Calculate R_{eq} with respect to $a-b$

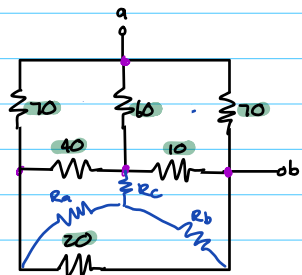
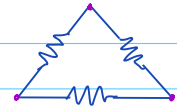


③ Calculate R_{eq} with respect to a-b



* All resistors are in parallel

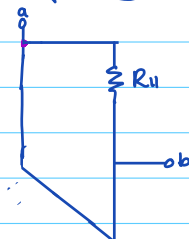
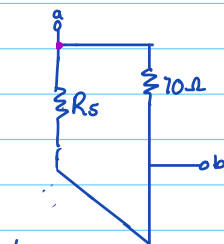
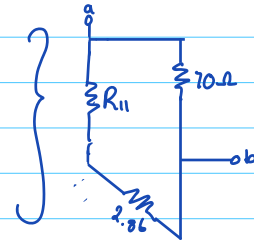
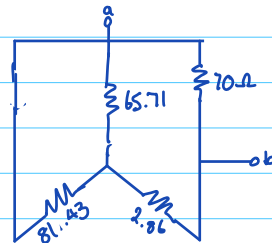
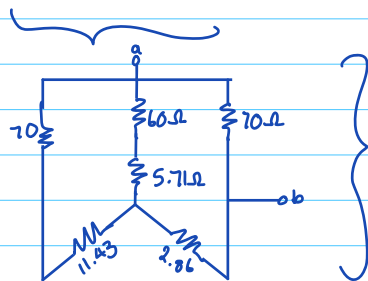
→ 3 "boxes" with resistors ∴ wye & delta



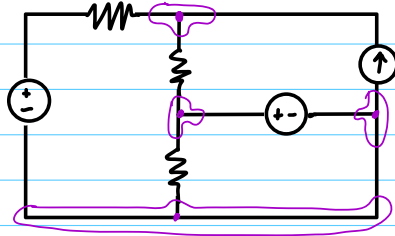
$$\therefore R_a = \frac{40 \times 20}{40 + 20 + 10} = 11.43 \Omega$$

$$R_b = \frac{20 \times 10}{40 + 20 + 10} = 2.857 \Omega$$

$$R_c = \frac{40 \times 10}{40 + 20 + 10} = 5.714 \Omega$$

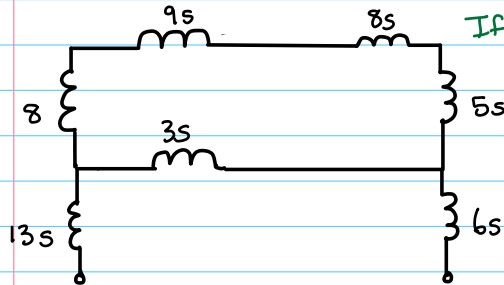


④ Determine nodes, branches, loops



$$\begin{aligned} n &= 4 \\ b &= 3 \\ b &= 6 \end{aligned}$$

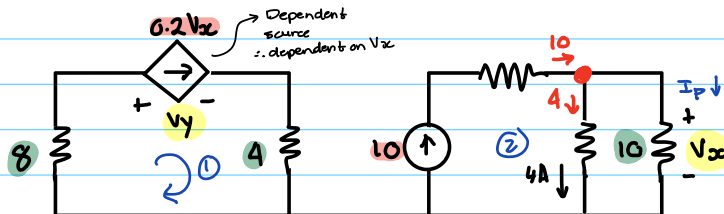
⑤ conductors - sm



If conductances in serie = we use parallel method to add them
If conductances in paralll = we use serie method to add them

$$\begin{aligned} 1. & \left(\frac{1}{8} + \frac{1}{9} + \frac{1}{8} + \frac{1}{5} \right)^{-1} = 1.827 \text{ S} \\ 2. & 1.827 + 3 = 4.827 \text{ S} \\ 3. & \left(\frac{1}{4.827} + \frac{1}{13} + \frac{1}{6} \right)^{-1} = 2.218 \text{ S} \end{aligned}$$

⑥



Loop 1

KVL: $I R_8 + V_y + I R_4 = 0$

$$\begin{aligned} I &= 0.2V_x \\ (0.2V_x)(8) + V_y + (0.2V_x)(4) &= 0 \\ 1.6V_x + 0.8V_x + V_y &= 0 \\ 2.4V_x + V_y &= 0 \end{aligned}$$

$$\begin{aligned} \therefore 2.4(60) + V_y &= 0 \\ V_y &= -144 \text{ V} \end{aligned}$$

Loop 2

$V_x = \pm [IR]$ $\rightarrow$ balance

$$\begin{aligned} \text{KCL: } 10 &= 4 + I_p \\ \therefore I_p &= 6 \\ \therefore V_x &= + (6)(10) \\ &= 60 \text{ V} \end{aligned}$$

Uncle Madel - Tutor Class 4

18 August 2022: 20:00 - 21:00

Revision: Matrices, Determinants, Cramer's Rules

* See WTW 158 Theme 6
WTW 164 Theme 1 Unit 1.2 - 1.8

EXAMPLES: Cramer's Rule

$$\overset{\Delta}{\begin{bmatrix} 2 & 4 \\ 6 & 3 \end{bmatrix}} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

① $\therefore \Delta = (2 \times 3) - (4 \times 6) = -18$

② • $\Delta_{I_1} = \begin{bmatrix} 5 & 4 \\ 7 & 3 \end{bmatrix}$

• $\Delta_{I_1} = (5 \times 3) - (7 \times 4) = -13$

• $I_1 = \frac{\Delta_{I_1}}{\Delta} = \frac{-13}{-18} = 0.722 \text{ A}$

③ • $\Delta_{I_2} = \begin{bmatrix} 2 & 5 \\ 6 & 7 \end{bmatrix}$

• $\Delta_{I_2} = (2 \times 7) - (6 \times 5) = -16$

• $I_2 = \frac{\Delta_{I_2}}{\Delta} = \frac{-16}{-18} = 0.89 \text{ A}$

1. Determine determinant of Δ

2. To determine I_1 :

- Substitute "answer-column" in I_1 column in Δ

- Determine new determinant Δ_{I_1}

- Calculate I_1 by $I_1 = \frac{\Delta_{I_1}}{\Delta}$

3. To determine I_2 :

- Substitute "answer-column" in I_2 column in Δ

- Determine new determinant Δ_{I_2}

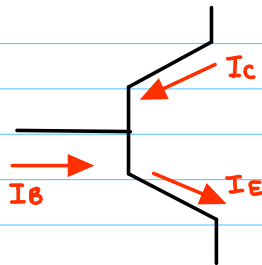
- Calculate I_2 by $I_2 = \frac{\Delta_{I_2}}{\Delta}$

Uncle Madel - Tutor Class 6

22 August 2022: 20:30 - 22:30

3.9 1. Transistors

currents

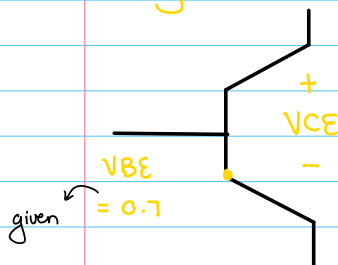


Calculate I_E first $\therefore$ write everything in terms of I_B

$$I_E = (1 + \beta) I_B$$

$$I_C = \beta \cdot I_B$$

voltages

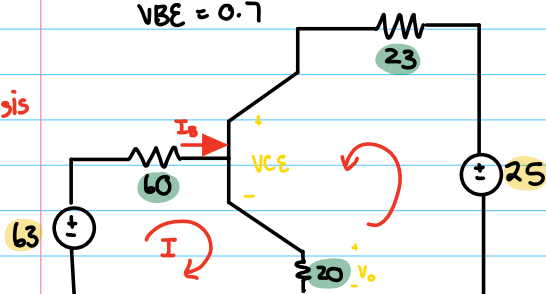


EXAMPLE 1

Given: $\beta = 99$

$$V_{BE} = 0.7$$

* KVL or
Nodal analysis



a) Calculate I_E :

$$\text{KVL: } -63 + 60 I_B + 0.7 + 20 I_E = 0$$

$$-63 + 60 I_B + 0.7 + 20 (1 + \beta) I_B = 0$$

$$\therefore I_B = 0.034$$

b) Calculate V_0 :

$$V_0 = \pm [I R]$$

$$= \pm [I_E R]$$

$$= (1 + \beta) (0.03) \times 20$$

↑

$$* I_E = (1 + \beta) I_B$$

$$V_0 = 60V$$

* With transistors: Change loop rotation
because we don't want loop to oppose
direction

c) Calculate V_{CE} :

$$\text{KVL: } -25 + 23 I_C + V_{CE} + 20 I_E = 0$$

$$-25 + 23 \beta I_B + V_{CE} + 20 (1 + \beta) I_B = 0$$

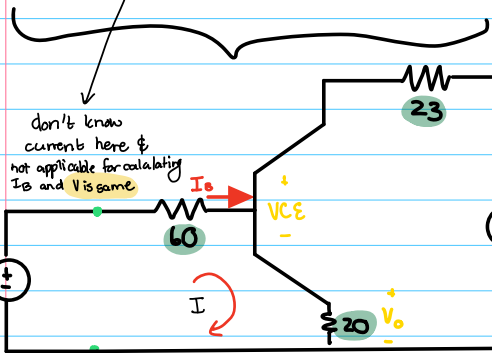
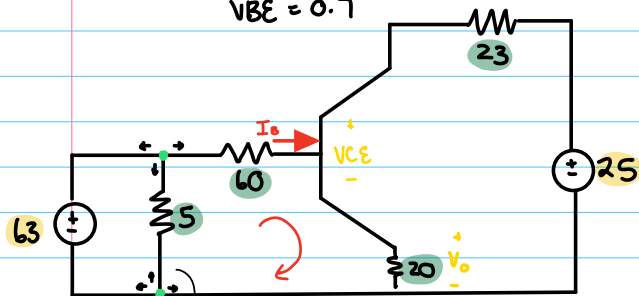
$$-25 + 23 (99)(0.03) + V_{CE} + 20 (1 + 99)(0.03) = 0$$

$$V_{CE} =$$

EXAMPLE 2

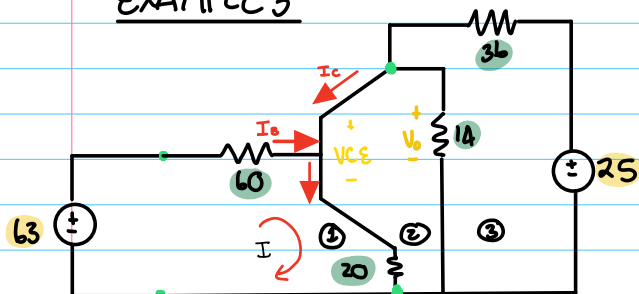
Given: $\beta = 99$

$V_{BE} = 0.7$



a) Calculate V_o

EXAMPLE 3



1. $I_B = 0.03$

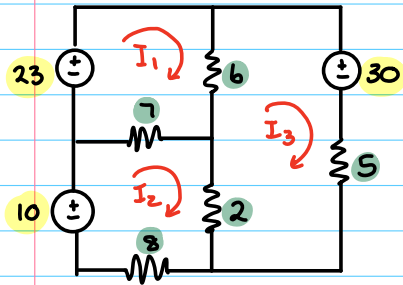
NB! Nodal analysis

$$1. \frac{V_o}{14} + \frac{V_o - 25}{36} + I_C = 0$$

$$\frac{V_o}{14} + \frac{V_o - 25}{36} + (99 \times 0.03) = 0$$

$$\therefore V_o =$$

EXAMPLE 4 : Inspection by mesh



$$\begin{array}{c} 1 \\ 2 \\ 3 \end{array} \begin{array}{c} 1 \quad 2 \quad 3 \\ \left[\begin{array}{ccc} +13 & -7 & -6 \\ -7 & +17 & -2 \\ -6 & -2 & +13 \end{array} \right] \end{array} \begin{array}{c} I_1 \\ I_2 \\ I_3 \end{array} = \begin{array}{c} \overset{?}{\left[\begin{array}{c} +23 \\ +10 \\ -30 \end{array} \right]} \end{array} \begin{array}{c} \left[\begin{array}{c} -23 \\ -10 \\ +30 \end{array} \right] \end{array}$$

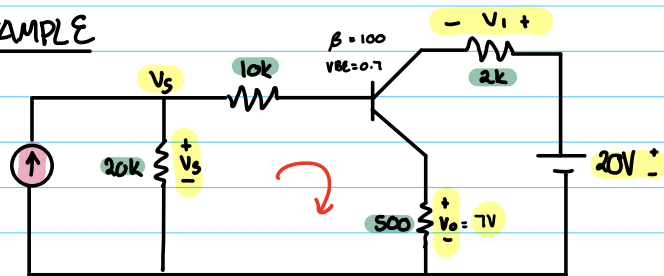
EXAMPLE 5 : Inspection by mesh

$$\begin{array}{c} 1 \\ 2 \\ 3 \end{array} \begin{array}{c} 1 \quad 2 \quad 3 \\ \left[\begin{array}{ccc} p & -7 & -6 \\ \beta & +17 & \alpha \\ -6 & -2 & +13 \end{array} \right] \end{array} \begin{array}{c} I_1 \\ I_2 \\ I_3 \end{array} = \begin{array}{c} \left[\begin{array}{c} k \\ 10 \\ -30 \end{array} \right] \end{array} \quad \begin{array}{l} \therefore k = 23 \\ \beta = -7 \\ \alpha = -2 \\ p = 13 \end{array}$$

Uncle Madel - Tutor Class 7

25 August 2022: 20:00 - 22:00

EXAMPLE



1. Since you know V_0 , then you can find I_B (your girlfriend)

$$V_0 = 500 I_E = 7 \text{ KVL}$$

$$I_E = \frac{7}{500}$$

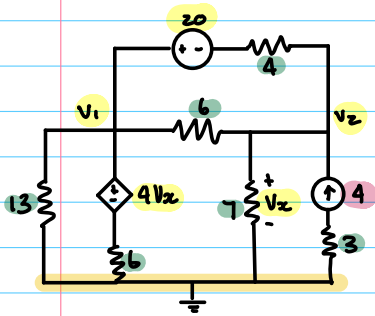
$$* I_E = (1 + \beta) I_B$$

$$I_B (1 + \beta) = \frac{7}{500}$$

$$I_B (1 + 100) = \frac{7}{500}$$

$$I_B = 140 \mu A$$

Inspection by nodal analysis



Case no. 1

Nodal analysis

$$\text{KCL: @ } V_1: \frac{V_1}{13} + \frac{V_1 - 4V_2}{6} + \frac{V_1 - 20 - V_2}{4} + \frac{V_1 - V_2}{6} = 0$$

$$\frac{V_1}{13} + \frac{V_1 - 4V_2}{6} + \frac{V_1 - 20 - V_2}{4} + \frac{V_1 - V_2}{6} = 0$$

$$* V_2 = V_2$$

$$\frac{V_1}{13} + \frac{V_1 - 4V_2}{6} + \frac{V_1 - 20 - V_2}{4} + \frac{V_1 - V_2}{6} = 0$$

$$\textcircled{1} \frac{103}{156} V_1 - \frac{13}{12} V_2 = 5$$

$$\text{KCL: @ } V_2: \frac{V_2}{7} + \frac{V_2 - V_1}{6} + \frac{V_2 - (-20) - V_1}{4} - 4 = 0$$

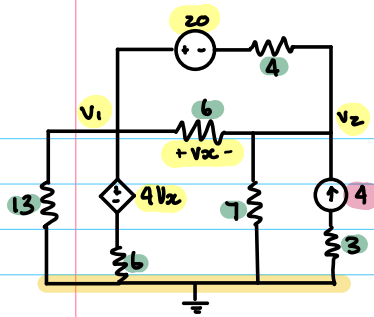
$$\textcircled{2} -\frac{5}{12} V_1 + \frac{47}{84} V_2 = -1$$

Put in calculator: Mode $\rightarrow$ EQN $\rightarrow$ $a_n X + b_n Y = c_n$

	V_1	V_2	constant
$\textcircled{1}$	$\frac{103}{156}$	$-\frac{13}{12}$	5
$\textcircled{2}$	$-\frac{5}{12}$	$\frac{47}{84}$	-1

Calculator $\therefore X = -20.916 = V_1$

$Y = -17.363 = V_2$



Case no. 2

Nodal analysis with supernode

$$\text{KCL: @ } V_1: \frac{V_1 - V_2}{6} + \frac{V_1 - 20 - V_2}{4} + \frac{V_1}{13} + \frac{V_1 - 4V_x}{6} = 0$$

$$* V_x = V_1 - V_2$$

$$\therefore \frac{V_1 - V_2}{6} + \frac{V_1 - 20 - V_2}{4} + \frac{V_1}{13} + \frac{V_1 - 4(V_1 - V_2)}{6} = 0$$

$$\text{KCL: @ } V_2: \frac{V_2}{7} + \frac{V_2 - V_1}{6} + \frac{V_2 - (-20) - V_1}{4} - 4 = 0$$

$$-\frac{5}{12} V_1 + \frac{47}{84} V_2 = -1$$

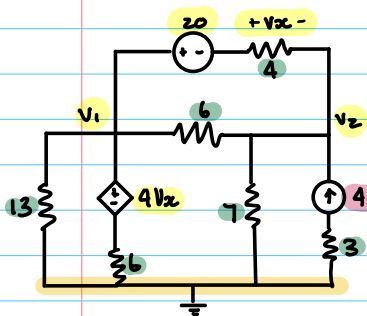
Put in calculator : Mode $\rightarrow$ EQN $\rightarrow$ $a_n X + b_n Y = C_n$

	V_1	V_2	constant
$\therefore$	① ()	()	()
	② $(-\frac{5}{12})$	$(\frac{47}{84})$	(-1)

Calculator $\therefore$ $X =$ $= V_1$
 $Y =$ $= V_2$

Case no. 3

Nodal analysis with Ohm's law



$$\text{KCL: @ } V_1: \frac{V_1}{13} + \frac{V_1 - 20 - V_2}{4} + \frac{V_1 - V_2}{6} + \frac{V_1 - 4V_x}{6} = 0$$

$$V_x = \pm I R$$

$$V_x = \left(\frac{V_1 - 20 - V_2}{4} \right) \times 4$$

$$V_x = V_1 - 20 - V_2$$

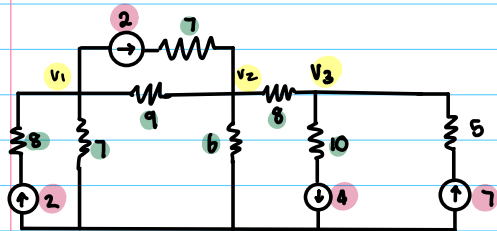
$$* \text{ or short cut: } V_1 - V_2 = 20 + V_x$$

$$\therefore V_1 - V_2 - 20 = V_x$$

$$\text{KCL: @ } V_2: \frac{V_2}{7} + \frac{V_2 - V_1}{6} + \frac{V_2 - (-20) - V_1}{4} - 4 = 0$$

$$-\frac{5}{12} V_1 + \frac{47}{84} V_2 = -1$$

EXAMPLE 2



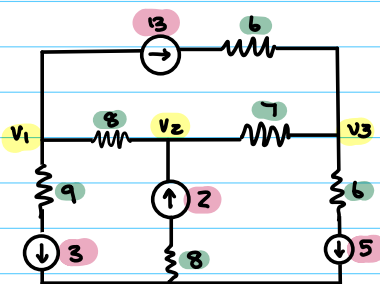
1. Convert $R \rightarrow G \therefore G = \frac{1}{R}$

INSPECTION with Conductances

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & \frac{1}{7} + \frac{1}{9} + \frac{1}{8} & -\frac{1}{9} + \frac{1}{7} & -0 \\ 2 & -\frac{1}{9} + \frac{1}{7} & \frac{1}{9} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} & -\frac{1}{8} \\ 3 & 0 & -\frac{1}{8} & \frac{1}{10} + \frac{1}{5} + \frac{1}{8} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 2-2 \\ 2 \\ -4+7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 0.522 & -0.254 & 0 \\ 2 & -0.254 & 0.546 & -0.125 \\ 3 & 0 & -0.125 & 0.425 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$$

EXAMPLE 3



INSPECTION with Conductances

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & \frac{1}{8} + \frac{1}{9} + \frac{1}{6} & -\frac{1}{8} & -\frac{1}{6} \\ 2 & -\frac{1}{8} & \frac{1}{8} + \frac{1}{6} + \frac{1}{7} & -\frac{1}{7} \\ 3 & -\frac{1}{6} & -\frac{1}{7} & \frac{1}{7} + \frac{1}{6} + \frac{1}{6} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -3-13 \\ +2 \\ -5+15 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 0.402 & -0.125 & -0.1667 \\ 2 & -0.125 & 0.3929 & -0.1429 \\ 3 & -0.1667 & -0.1429 & 0.4762 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -16 \\ 2 \\ 8 \end{bmatrix}$$

Uncle Mabel - Tutor Class 8

10 September 2022: 20:00 - 22:00

CHAPTER 4

- Linearity (later)
- Superposition

Superposition

① Close/Remove independent sources

- * close voltage source that is not in use with line
- * remove current source that is not in use

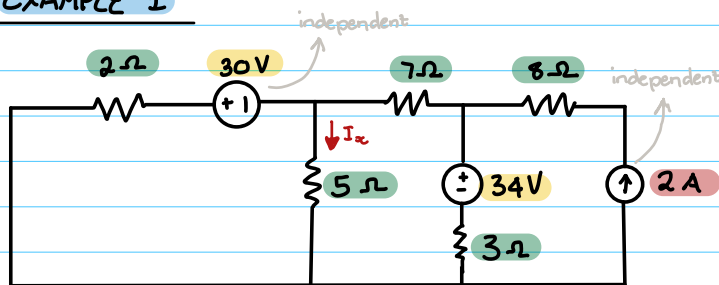
② Use nodal analysis (refer to chapter 3, class 3)

1. Identify a node (where 3/more lines connecting)
2. Assign variable(s) & 1 node must sacrifice (has to be ground)
3. — (will do later)
4. Assume current is leaving the node that we are dealing with
5. Apply node analysis

• Determine the contribution of ∞ to y

• Calculate the contribution of y due ∞

EXAMPLE 1



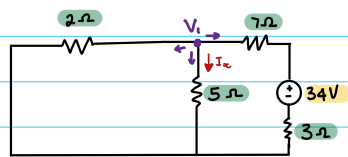
1. Calculate the contribution of I_x due to 34V → use only this source if asked

① Close the 30V with line

- Remove 2A



② Nodal analysis



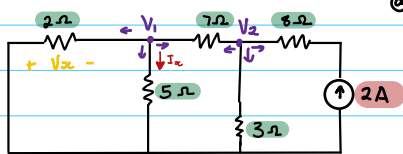
$$\begin{aligned} \cdot \frac{V_1}{2} + \frac{V_1}{5} + \frac{V_1 - (+34)}{10} &= 0 \\ 5V_1 + 2V_1 + V_1 - 34 &= 0 \\ 8V_1 &= 34 \\ V_1 &= 4.25V \end{aligned}$$

$$\begin{aligned} \cdot \frac{V_1}{5} &= I_x \\ I_x &= 0.85A \\ & (= 850mA) \end{aligned}$$

2. Calculate the contribution of 2A to V_{oc}

① Close the 30V and 34V with line

② Nodal analysis



$$\begin{aligned} @V_1: \frac{V_1}{2} + \frac{V_1}{5} + \frac{V_1 - V_2}{7} &= 0 \\ 35V_1 + 14V_1 + 10V_1 - 10V_2 &= 0 \\ 59V_1 - 10V_2 &= 0 \end{aligned}$$

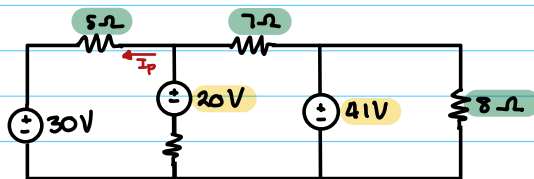
$$\begin{aligned} @V_2: \frac{V_2 - V_1}{7} + \frac{V_2}{3} - 2 &= 0 \\ 3V_2 - 3V_1 + 7V_2 - 42 &= 0 \\ -3V_1 + 10V_2 &= 42 \end{aligned}$$

$$\therefore V_1 = 0.75V$$

$$V_2 = 4.425V$$

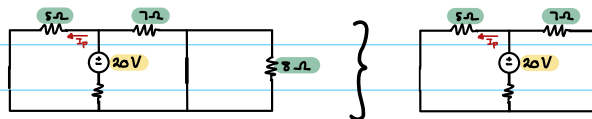
$$\text{NB! } V_x = -V_1 = -0.75V$$

EXAMPLE 2



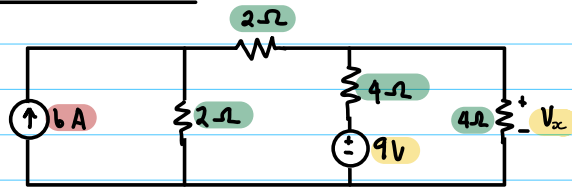
1. Calculate the contribution of 20V to I_p

① Close the 41V and 30V with line



② Nodal analysis

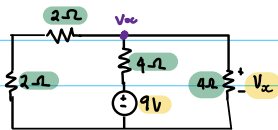
EXAMPLE 3



1. Calculate the contribution of V_x due to 9V

① Remove 6A

② Nodal analysis



$$\frac{V_x}{4} + \frac{V_x - 9}{4} + \frac{V_x}{4} = 0$$

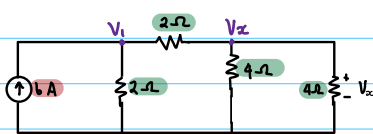
$$3V_x = 9$$

$$V_x = 3V$$

2. Calculate the contribution of V_x due to 6A

① Close 9V with line

② Nodal analysis



$$\text{@ } V_1: \frac{V_1 - V_x}{2} + \frac{V_1}{2} - 6 = 0$$

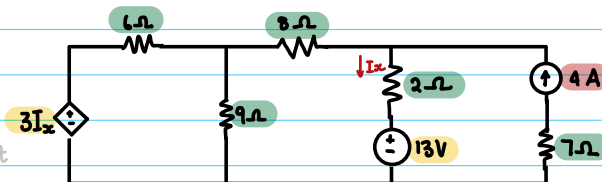
$$2V_1 - V_x = 12$$

$$\text{@ } V_x: \frac{V_x - V_1}{2} + \frac{V_x}{4} + \frac{V_x}{4} = 0$$

$$4V_x - 2V_1 = 0$$

$$2V_x - V_1 = 0$$

EXAMPLE 4

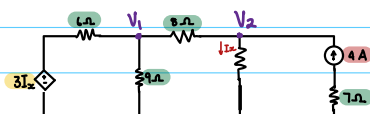


NB!
LEAVE dependent
sources!

1. Calculate the contribution of I_x due to 4A

① Close 13V with line

② Nodal analysis

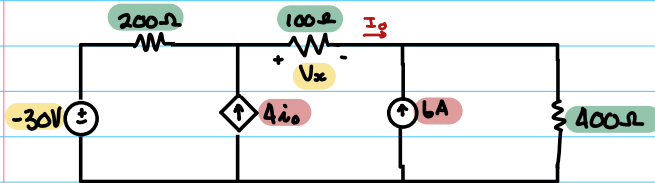


$$\frac{V_1 - 3I_x}{6} + \frac{V_1 - V_2}{8} + \frac{V_1}{9} = 0$$

$$* I_x = \frac{V_2}{2}$$

$$\frac{V_1 - (3/2)V_2}{6} + \frac{V_1 - V_2}{8} + \frac{V_1}{9} = 0$$

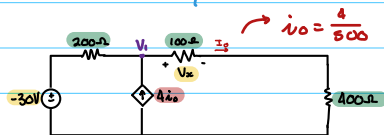
EXAMPLE 5



1. Calculate the contribution of $-30V$ due to V_x

① Remove 6A

② Nodal analysis



$$\frac{V_1 - (-30)}{200} + \frac{V_1}{500} - 4\left(\frac{V_1}{500}\right) = 0$$

$$V_1 = 150V$$

$$V_x = \pm [IR]$$

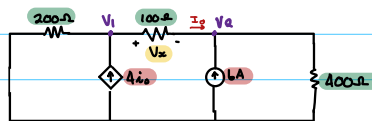
$$= + \left[\frac{V_1}{500} \times 100 \right]$$

$$V_x = 30V$$

2. Calculate the contribution of $6A$ to V_x

① Close $-30V$ with line

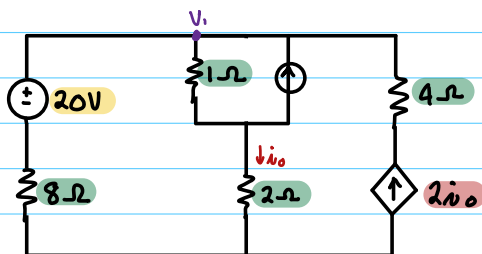
② Nodal analysis



$$@V_1: \frac{V_1}{200} - 4\left(\frac{V_1 - V_2}{100}\right) + \frac{V_1 - V_2}{100} = 0$$

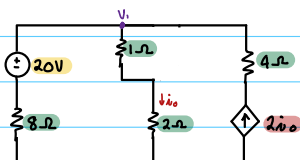
$$@V_2: \frac{V_2 - V_1}{100} + \frac{V_2}{400} - 6 = 0$$

EXAMPLE 6



① Remove current source

② Nodal analysis



$$\frac{V_1 - 20}{8} + \frac{V_1}{3} - 2\left(\frac{V_1}{3}\right) = 0$$

$$* i_o = \frac{V_1}{3}$$